

Divyaansh Seth

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EDUCATION

University of Massachusetts Amherst

Sep 2023 – May 2027

B.S. Computer Science and Mathematics; Minor: Business

GPA: 3.744/4.0

- **Honors:** Dean's List, Chancellor's Scholar
- **Relevant Coursework:** Computer Systems Principles, Introduction to Algorithms, Data Structures, C Programming, Reasoning Under Uncertainty, Mathematical Foundations of Statistics & Data Science, Machine Learning, Probability & Statistics, Linear Algebra, Ordinary Differential Equations

TECHNICAL SKILLS

Languages: C++, Python, C, Java, SQL

Systems: Linux/Unix, Concurrency, Multithreading, AsyncIO, Synchronization, Sockets, Network Programming, Distributed Systems, Memory Management, RAII, File I/O

Tools & Infrastructure: Git, Bash, Make, CMake, Docker, Podman, gRPC, Kafka, Redis, PostgreSQL, GitHub Actions, CI/CD

Performance & Quantitative: Latency Benchmarking, OpenTelemetry, Distributed Tracing, NumPy, Pandas, SciPy, Backtesting, Statistical Modeling, Risk Analytics

EXPERIENCE

Siemens Energy

Jun 2026 – Present

Software Engineering / AI Platform Intern

Gurugram, India

- Owned the end-to-end conversion of a Jupyter prototype into a distributed 3-engine Python/FastAPI application in 1 week, defining service boundaries, parallelizing two long-running processing paths, and deploying containerized API/worker workloads.
- Profiled stage-level latency across a distributed processing pipeline and optimized batching, parallel execution, token usage, database queries, and I/O; replaced an unnecessary model stage with deterministic JSON processing, reducing end-to-end runtime **13% from 6m44s to 5m50s**.
- Diagnosed repeated execution of >5-minute asynchronous jobs by live-tracing local and containerized API/worker processes alongside Service Bus queue state; identified competing consumers and message-lock expiration, then implemented isolated test queues and automatic lock renewal to prevent duplicate processing.
- Engineered asynchronous backend infrastructure supporting **20–25 concurrent requests** with **3–10 autoscaled worker replicas**, 2-vCPU/4-GiB container limits, distributed tracing across 9 services, and priority routing across 4 service endpoints in 2 regions.
- Hardened worker reliability with duplicate-job detection, redelivery protection, terminal-state safeguards, correlation-based observability, and **45 passing unit tests** for production handover.

Deloitte Touche Tohmatsu India LLP

Jul 2025 – Aug 2025

Summer Intern – AI Engineer

Gurugram, India

- Engineered a DINOv2 + FAISS HNSW retrieval pipeline over **2,000+ embryo images**, achieving **84% accuracy** against embryologist annotations; integrated morphology features into ensemble predictive models using XGBoost and Random Forest.
- Applied Boruta feature selection and inference optimization across 30+ clinical variables, achieving sub-2-second inference and improving prediction calibration across age groups by **18%**.

National Informatics Centre, Government of India

Jun 2025 – Jul 2025

Summer Intern – AI Engineer

New Delhi, India

- Built Redis-backed ingestion/session microservices for multilingual retrieval using Qwen3 embeddings, Milvus vector search, and Mistral LLM, reducing irrelevant or outdated responses by **30%**.

PROJECTS

Low-Latency Limit Order Book & Matching Engine

Aug 2026 – Present

C++20, STL, Multithreading, CMake, Linux/Unix

- Built a C++20 matching engine with price-time priority, fixed-point integer prices, limit/market orders, partial/full fills, cancel/modify operations, duplicate-ID rejection, aggregate depth tracking, and **17 always-on correctness tests**.
- Benchmarked pre-generated balanced workloads at **6.27M events/sec** with **0.42 μ s p99 raw engine-processing latency** across 1M events; separately measured a bounded producer/consumer pipeline at **1.93M events/sec** and isolated queue wait as the dominant source of tail latency.

Algorithmic Trading Strategy Backtester

Apr 2025 – Jun 2025

Python, NumPy, Pandas

- Built a modular backtesting engine with pluggable strategy interfaces, historical OHLCV ingestion/caching, iterative position and cash accounting, commissions, slippage, and risk analytics including Sharpe ratio, maximum drawdown, VaR, and CVaR.